14.8 Writing and Solving One-Step Equations in One Variable

Lesson Introduction

Benchmarks	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.	Learning Targets	• I can write and solve one-step equations in one variable within a real-world context using multiplication and division. • I can solve one-step equations in one variable using addition and subtraction.
Benchmark Coherence	**Building On** MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.	**Working Toward** MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
Essential Questions	• How do you write and solve multiplication and division one-step equations from a real-world context? • How can you check your solution to an equation is reasonable?		
Lesson Narrative	In this lesson, students connect one-step multiplication and division equations to real-world situations. Students are first presented with real-world situations and find the equation that models each situation. Students are given real-world situations where they have to write an equation to represent the situation and solve the equation. The lesson ends with a mix review of solving one-step equations.		

Component Summary

14.8.1 Warm-Up (~5 min.)	Solve a puzzle of one-step equations *(SE p. 379)*	14.8.4 Guided Practice (10 min.)	Write and solve one-step real-world multiplication and division equations *(SE p. 381)*
14.8.2 Collaboration (5 min.)	Collaboratively match real-world situations to one-step equations *(SE p. 379)*	14.8.5 Try It (10 min.)	Mixed review writing and solving one-step equations *(SE p. 382)*
14.8.3 Guided Instruction (15 min.)	Write and solve one-step real-world multiplication and division equations *(SE p. 380)*	14.8.6 Wrap-Up (~5 min.)	Self-assessment of mastery of writing and solving one-step equations *(SE p. 383)*

Resources

Glossary	Materials	Additional Preparation
Key Terms: N/A Additional Terms: N/A	• (Optional) Chart paper • (Optional) Markers	• Consider modifying the context of some questions to connect them to contexts students in your class are interested in. • Consider including an example problem for each learning target in the 14.8.6 Wrap-Up to help students accurately self-assess their mastery of the target.

Teacher Tips

Common Misconceptions

- Students may not perform the inverse operation on both sides of the equation.
- Students may believe each situation can only be represented by one equation, therefore struggling to identify equivalent equations.

Things to Consider

- Have students continue to create their own word problems using different equations and verbalize what the variable is in their problem.
- If students are having problems with creating their own equations for the questions, have them draw a visual model that represents the problem and then translate the visual model into an equation.
- If time is an issue, then consider assigning 14.8.5 Try It for homework.
- Students who can do the mathematics in their head may struggle with the purpose of writing an equation. These students may also write the equations as $x = 24 \div 8$ because that is how they see the problem. Their equation could be mathematically accurate and could not be considered wrong. For these students it may be best to ask that they write two equivalent equations. Forcing the student to write $8x = 24$ cannot be justified mathematically since their equation is equivalent. It may be helpful to have state the expected form of the equations to be $px = q$ or $\frac{x}{p} = q$.

Literacy and Language Routines

Think-Pair-Share

In 14.8.2 Collaboration, ask students to initially determine the matches that they believe are correct independently. Then have students discuss their answers and justifications with a partner. These discussions will build student confidence and understanding.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.7.1: Real-World Contexts

In 14.8.3 Guided Instruction, students write and solve one-step multiplication and division equations based on real-world situations. These questions will help students develop an appreciation of the utility of math. Students will be increasingly more engaged if they are connected or interested in the context.

Differentiate Instruction

The scaffolding support of visual models have been used throughout this unit but are not prominent in this lesson. Some students may still require the support of the visual models as they work to solve one-step multiplication and division equations. Encourage students who may benefit from additional visual entry points to draw bar models before solving. Students may benefit from the additional kinesthetic entry point of algebra tiles as well. Consider providing these scaffolds during the 14.8.3 Guided Instruction and 14.8.4 Guided Practice portions of this lesson and releasing students to solve the rest of the problems algebraically if they can.

Lesson Notes:

14.8.1 Warm-Up: One-Step Equations

Component Overview: *Students create true equations by filling in the empty boxes using the digits one through nine.*

14.8.1 Warm-Up: One-Step Equations

1. Fill in each blue box to make the equations true. Use only digits 1 to 9, at most one time each.

$$\boxed{4} + a = \boxed{9}$$

$$\boxed{3}\, b = \boxed{6}$$

$$c - \boxed{7} = \boxed{1}$$

$$a = \boxed{5},\ b = \boxed{2},\ c = \boxed{8}$$

Consider having students share their strategies for filling in the boxes so that each equation is true.

14.8.2 Collaboration: Relating Real-World Contexts to Equations

Component Overview: *Students collaboratively match real-world situations to one-step equations.*

14.8.2 Collaboration: Relating Real-World Contexts to Equations

1. Four equations are shown.

$$4x = 32 \qquad 32x = 4$$
$$\frac{x}{32} = 4 \qquad \frac{x}{4} = 32$$

Discuss with your partner which of the equations correctly represents each situation below, where x represents the unknown quantity. Then, write the matching equation in the space provided.
Answers will vary.

Situation 1	Situation 2
At a restaurant, 4 friends decided to split the bill evenly. If each person paid \$32, what was the total bill?	A manatee has a total of 32 teeth. If a manatee has 4 rows of teeth with the same number of teeth in each row, how many teeth are in each row?
Equation: $\frac{x}{4} = 32$	**Equation:** $4x = 32$

Think-Pair-Share
Consider structuring this activity with a Think-Pair-Share routine.

- *Think: Students independently read each situation and determine the matching equation.*
- *Pair: Students turn to a partner to justify answers and discuss any differences.*
- *Share: Prompt students to share whole group any reasoning from their partner that helped to change their approach or answer.*

14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division

Component Overview: *In 14.8.3 Guided Instruction, students begin by creating a bar model to represent a real-world scenario. Students then create an equation to represent the model and use it to solve for the unknown variable, explaining what the solution means in the context of the original problem. Students then work with a partner to determine which equations accurately represent another scenario and choose an equation to solve for the unknown. Students then check their work using a different equation and explain their solution in the context of the original problem.*

14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division

1. A town's total budget for firefighters' wages is $\$320{,}000$. Wages are calculated at $\$40{,}000$ per firefighter.

   a. Draw a bar model that can be used to determine how many firefighters the town can afford to employ.
   Answers will vary.

   b. Write an equation that can be used to find w, the number of firefighters the town can employ.
   $40{,}000w = 320{,}000$

   c. What is the value of w?
   $w = 8$

   d. Explain what the solution means in the context of the problem.
   Answers will vary. The town can hire 8 firefighters.

Remind students that they can create visual models for all questions (even if the directions do not require them to). Also, model how to check the solution and remind students to check their solutions.

MA.K12.MTR.7.1: Real-World Contexts
Students write and solve one-step multiplication and division equations based on real-world situations. These problems will help students develop an appreciation of the utility of math while providing meaningful engagement for making sense in familiar contexts.

Consider allowing students to adjust the context based on their own lived experiences. For example, question 1 could be adjusted to school budgets and spending per student for meaningful conversations on equitable school funding, or question 2 could be adjusted to be music samples and recording studios in town.

14.8.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division (continued)

2. There are 4 flooring companies submitting bids on a new housing development. Each company has 11 different flooring samples.

   Dino said, "To find the total number of samples, s, being submitted, set up the equation $s = 4 \cdot 11$." Dino's partner Callahan shared that he used the equation $\frac{s}{4} = 11$ to find the total number of samples submitted.

   Bristol shouted across the classroom that she wrote the equation, $\frac{s}{11} = 4$.

   a. Discuss with your partner which student — Dino, Callahan, or Bristol — wrote the correct equation to solve for the total number of samples submitted.
   Answers will vary. All three wrote an equation that can be solved to find the number of samples.

   b. Solve one of the equations and then use a different equation to check your work.
   Answers will vary.
   Dino: $s = 4 \cdot 11$; $s = 44$
   Callahan: $\frac{s}{4} = 11$; $\frac{44}{4} = 11$
   Bristol: $\frac{s}{11} = 4$; $\frac{44}{11} = 4$

   c. Explain how the answer makes sense in terms of the context.
   Answers will vary. There are 44 samples. Multiplication and division are inverse operations therefore all three students are correct. Four stores times 11 samples equals 44.

If students struggle to express themselves with written responses, consider providing sentence stems or answer choices. This will enable students to demonstrate their understanding.

Students may struggle to understand that all three equations are correct and can be used to solve. Encourage students to think flexibly about the equations. To further scaffold, ask questions like:

- *What does the variable s represent in this problem?*
- *Is there only one way to solve for s based on the problem?*
- *What do you notice about each of the equations?*

14.8.4 Guided Practice: Solving Multiplication and Division Equations for Real-World Contexts

Component Overview: *Students create related multiplication and division equations to represent two real-world situations. Students then choose an equation to solve for the unknown variable and explain how the multiplication and division equations demonstrate inverse operations.*

14.8.4 Guided Practice: Solving Multiplication and Division Equations for Real-World Contexts

For each situation, write an equation that can be used to determine the unknown value. Then, solve the equation. Be sure to check your solution.

1. It takes 5 yards of fabric to make a Native American ceremonial dress.

   a. Write a division equation and a multiplication equation that can be used to determine how many yards, d, of fabric are needed to make 17 ceremonial dresses.

Division Equation	Multiplication Equation
$\frac{d}{5} = 17$	$5 \cdot 17 = d$

   b. Solve either of the equations for d. Use the other equation to check your work. $d = 85$

2. Drake is selling candy bars for a school fundraiser. Each candy bar sells for $\$3.00$.

   a. Write a division equation and a multiplication equation that can be used to determine how many candy bars, c, he sold if he raised $\$399$.

Division Equation	Multiplication Equation
$\frac{399}{3} = c$	$3c = 399$

   b. Solve either of the equations for c. Use the other equation to check your work.
   $c = 133$

3. Discuss with your partner how creating the two different equations in Questions 1 and 2 demonstrate inverse relationships. *Answers will vary. Creating two different equations shows inverse relationships by demonstrating how the operations undo each other to result in the same answer.*

Consider allowing students to reference a multiplication table while writing and solving the equations. It may be beneficial to encourage students to create fact families to help create the equations.

Students may have trouble articulating how the equations demonstrate inverse relationships. Circulate as students discuss question 3 with their partners. If you notice students struggling, ask questions to provide further scaffolding:

- *How are the equations similar? How are they different?*
- *What makes operations inverses?*
- *How do the equations show this relationship?*

14.8.5 Try It: Mixed Review

Component Overview: *Students engage in a mixed review involving writing and solving one-step equations.*

14.8.5 Try It: Mixed Review

Select all of the equations that represent each situation.

1. One Valencia orange tree can produce 275 oranges. A crate contains 25 oranges.

   a. How many crates can one Valencia orange tree fill up? Select all that apply.

   - ☐ $\frac{x}{25} = 275$
   - ☐ $275 = 25x$
   - ☐ $275x = 25$
   - ☐ $\frac{x}{275} = 25$
   - ☐ $\frac{275}{x} = 25$
   - ☐ $\frac{275}{25} = x$

   b. Pick one of the equations you selected and solve the problem. Show your reasoning.
   Answers will vary.
   $x = 11$

2. Aliyah and Tiana want to bake 3 carrot cakes. Each carrot cake needs 2 cups of salted butter.

   a. How much butter will they need to bake all 3 cakes? Select all that apply.

   - ☐ $\frac{x}{2} = 3$
   - ☐ $2x = 3$
   - ☐ $2(3) = x$
   - ☐ $3x = 2$
   - ☐ $6x = 2$
   - ☐ $\frac{x}{3} = 2$
   - ☐ $\frac{x}{3} = 6$

   b. Select one of the equations you selected and solve the problem. Show your reasoning.
   Answers will vary.
   $x = 6$

Consider creating and displaying an anchor chart with instructions, examples, and visual models for students to reference as they complete these questions independently.

If students struggle to solve equations algebraically, provide them with concrete manipulatives or tell them to draw visual models. Additionally, if students forget to balance their equations, explain that to keep the equation balanced, any operation performed on one side of the equals sign must be performed on the other side.

14.8.5 Try It: Mixed Review (continued)

3. Solve each equation for the variable. Verify your answers.

a. $5x = 230$ $\frac{5x}{5} = \frac{230}{5}$ $x = 46$	b. $y + 4 = -2$ $y + 4 - 4 = -2 - 4$ $y = -6$
c. $-16 + d = -26$ $-16 + 16 + d$ $= -26 + 16$ $d = -10$	d. $r - 96 = 103$ $r - 96 + 96 = 103 + 96$ $r = 199$
e. $\frac{k}{4} = -51$ $\frac{k}{4}(4) = -51(4)$ $k = -204$	f. $-156 = -13w$ $\frac{-156}{-13} = \frac{-13w}{-13}$ $w = 12$

Remind students to check their solutions. If students finish early, ask them to write real-world problems that can be represented by the equations.

14.8.6 Wrap-Up: Reflection

Component Overview: *Students self-assess their mastery of writing and solving one-step equations.*

14.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning objective.
   Answers will vary.

I need help. I got it. I could help someone else.

Determine the value of an unknown decimal or fraction in an equation.	
Translate a written description into an algebraic expression.	
Write a description within a real-world context given an algebraic expression.	
Write and solve one-step equations in one variable using addition and subtraction.	
Write and solve one-step equations in one variable using multiplication and division.	
Determine the value of an unknown decimal or fraction in an equation.	

Consider including an example problem for each item to help students accurately self-assess their mastery of the target.